OpenShell

Purpose and PoC objective

- NVIDIA OpenShell is designed to isolate AI agent workloads with kernel and network policies.

  Evidence: report.md, evaluation.md

- Build three Rust runtime components as UBI9 images and test control-plane access and sandbox behavior on OpenShift.

  Evidence: report.md

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Testing strategy

- Five planned checks cover health, metrics, mTLS API, sandbox lifecycle and network policy.

  Evidence: plan.md

- A reachable out-of-policy control was added to the harness; its scenario remained skipped. Filesystem policy was not tested.

  Evidence: report.md

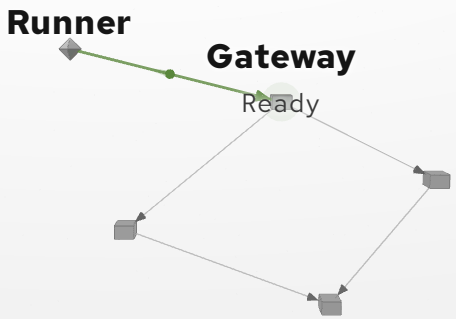


Recorded setup

Control-plane checks passed

CONTROL PLANE

3 pass




```
> Control plane          [+] 3 pass
  Lifecycle              [x] error
  Network policy         [-] skip
```

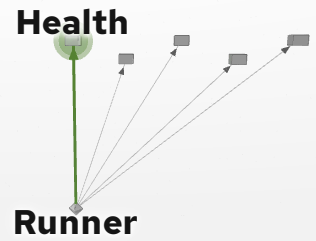
Animation illustrates archived evidence; playback timing is illustrative.

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Report-based executive summary

How we tested

Health: pass




```
> Health                    [+] pass
Metrics                     [+] pass
mTLS API                    [+] pass
Lifecycle                   [x] error
Network policy              [-] skip
```

HEALTH

pass

- Final capture: 3 pass, 1 error, 1 skip. Health, metrics and mTLS Health/list passed; prior attempts remain in history.
  Evidence: test-results.json, attempt-index.json
- Sandbox readiness timed out after 240s with an IBM disk-attachment quota. Network-policy validation was skipped.
  Evidence: report.md, test-results.json

Animation illustrates archived evidence; playback timing is illustrative.

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Report-based executive summary

Validation results

3	0	1	1	0
pass	fail	error	skip	incomplete

- No network/filesystem isolation, CNI, load/HA or wider Kubernetes portability validation.
- mTLS used a local-dev principal; production OIDC was not tested.
- Storage blocked sandbox validation. Operator-assisted PoC; production readiness is unproven.



OpenShift and Kubernetes conclusions

- Gateway and supervisor ran on OpenShift 4.21.8 under restricted-v2. All three images were source-built with documented patches.

  Evidence: report.md

- Operators assisted the build, RBAC and same-zone placement. Security, HA and wider Kubernetes portability remain unverified.

  Evidence: report.md

- No network/filesystem isolation, CNI, load/HA or wider Kubernetes portability validation.
- mTLS used a local-dev principal; production OIDC was not tested.
- Storage blocked sandbox validation. Operator-assisted PoC; production readiness is unproven.

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